

Education and Research Experiences

Graduate: MIT CSAIL

Advisor: Prof. Richard Ryan Williams

Sep 2023 -

Cambridge, Massachusetts, USA

Undergraduate: Tsinghua University

Institute for Interdisciplinary Information Sciences (Yao Class)

Sep 2019 - Jul 2023

Beijing, China

Research Internship: NTT Research

Advisor: Abhishek Jain, Senior Scientist

Jun - Aug 2025 & 2026

Sunnyvale, California, USA

Research Internship: Shanghai Qi Zhi Institute

Advisor: Prof. Yilei Chen. Research Project: Computational Complexity of Explicit Constructions.

Aug 2022 - Sep 2022

Shanghai, China

Research Internship: The University of Warwick

Advisor: Prof. Igor Carboni Oliveira. Research Project: Unprovability of Complexity Lower Bounds.

Mar 2022 - Jul 2022

Coventry, UK

Research Interests

- **Computational Complexity** (with an emphasis on circuit lower bounds)
- **Proof complexity** (and meta-mathematics of complexity theory)
- **Cryptography** (and its connection to complexity theory)

Publications

(All authors are with equal contribution and listed in the alphabetic order by default)

Topics: CC Computational Complexity PF Proof & Meta-Complexity CR Cryptography

Conferences

STOC 2026 CR PF SNARGs for NP from Unprovability of Mathematical Theorems (Or: How to use the simplicity of cryptographic reasoning).

Yao-Ching Hsieh (University of Washington), Abhishek Jain (NTT Research and John Hopkins), [Jiatu Li](#), Surya Mathialagan (NTT Research)

STOC 2026 PF CC A Theory for Probabilistic Polynomial-Time Reasoning.

Lijie Chen (UC Berkeley), [Jiatu Li](#), Igor C. Oliveira (University of Warwick), Ryan Williams (MIT)

ITCS 2026 CC Identity Testing for Circuits with Exponentiation Gates. [\[Manuscript\]](#)

[Jiatu Li](#), Mengdi Wu (CMU)

STOC 2025 CC Maximum Circuit Lower Bounds for Exponential-time Arthur Merlin. [\[Conference Version\]](#) [\[Full Version\]](#)

Lijie Chen (UC Berkeley), [Jiatu Li](#), Jingxun Liang (CMU)

STOC 2025 CC The Structure of Catalytic Space: Capturing Randomness and Time via Compression.

[\[Conference Version\]](#) [\[Full Version\]](#)

James Cook, [Jiatu Li](#), Ian Mertz (University of Warwick), Edward Pyne (MIT)

FOCS 2024 PF Reverse Mathematics of Complexity Lower Bounds. [\[Conference Version\]](#) [\[Full Version\]](#)

Lijie Chen (UC Berkeley), [Jiatu Li](#), Igor C. Oliveira (University of Warwick)

Invited to **Special Issue of SICOMP**; see [\[Journal Version\]](#). Also see [Quanta Magazine Coverage](#).

FOCS 2024 CC Distinguishing, Predicting, and Certifying: On the Long Reach of Partial Notions of Pseudorandomness.

[Conference Version](#) [\[Full Version\]](#)

[Jiatu Li](#), Edward Pyne (MIT), Roei Tell (University of Toronto)

STOC 2024 CC CR Hardness of Range Avoidance and Remote Point for Restricted Circuits via Cryptography.

[\[Conference Version\]](#) [\[Full Version\]](#)

Yilei Chen (Tsinghua University), [Jiatu Li](#)

STOC 2023 PF Unprovability of Strong Complexity Lower Bounds in Bounded Arithmetic.

[\[Conference Version\]](#) [\[Full Version\]](#)

[Jiatu Li](#), Igor C. Oliveira (University of Warwick).

STOC 2023 **CC** **PF** **CR** Indistinguishability Obfuscation, Range Avoidance, and Bounded Arithmetic.

[\[Conference Version\]](#) [\[Full Version\]](#)

Rahul Ilango (MIT), [Jiatu Li](#), Ryan Williams (MIT).

STOC 2023 **CC** Range Avoidance, Remote Point, and Hard Partial Truth Table via Satisfying-Pairs Algorithms.

[\[Conference Version\]](#) [\[Full Version\]](#)

Yeyuan Chen (Xi'an Jiaotong University), Yizhi Huang (Tsinghua University), [Jiatu Li](#), Hanlin Ren (University of Oxford).

CCC 2022 **CC** **CR** Extremely Efficient Constructions of Hash Functions, with Applications to Hardness Magnification and PRFs. [\[Conference Version\]](#) [\[Full Version\]](#)

Lijie Chen (MIT), [Jiatu Li](#), Tianqi Yang (Tsinghua University)

STOC 2022 **CC** **CR** The Exact Complexity of Pseudorandom Functions and the Black-Box Natural Proof Barrier for Bootstrapping Results in Computational Complexity. [\[Conference Version\]](#) [\[Full Version\]](#)

Zhiyuan Fan (Tsinghua University), [Jiatu Li](#), Tianqi Yang (Tsinghua University)

Best Student Paper co-winner. Invited to **Special Issue of SICOMP**.

STOC 2022 **CC** $3.1n - o(n)$ Circuit Lower Bounds for Explicit Functions. [\[Conference Version\]](#) [\[Full Version\]](#)

[Jiatu Li](#), Tianqi Yang (Tsinghua University).

Journals

LMCS 2025 **PF** On the Unprovability of Circuit Size Bounds in Intuitionistic S^1_2 . [\[Journal Version\]](#)

Lijie Chen (UC Berkeley), [Jiatu Li](#), Igor C. Oliveira (University of Warwick)

Manuscripts

In Submission **PF** On the Time Complexity of Feasible Proofs.

[Jiatu Li](#)

In Submission **CR** **PF** Overcoming Padding in Cryptography via the Hardness of Certifying Random Strings.

Yao-Ching Hsieh (University of Washington), Abhishek Jain (NTT Research and John Hopkins), [Jiatu Li](#), Surya Mathialagan (NTT Research)

Professional Service

- Conference Review: CCC (2022, 2024, 2025), STOC (2024, 2025), FOCS (2024, 2025, 2026), ITCS (2025, 2026)
- Journal Review: Forum Math. Sigma

Selected Awards and Fellowships

- MIT Schwarzman College of Computing PhD Fellowship, 2026–2027
- Jane Street Graduate Fellowship Honorable Mentions, 2025 and 2026.
- MIT Akamai Presidential Fellowship, 2023
- STOC 2022 Danny Lewin Best Student Paper Award
- Yao Award 1st Prize (1/62), IIIS, Tsinghua University, 2022

Selected Academic Talks

- Bounded Arithmetic Meets Probability, and Applications in Cryptography. [CSDM Seminar at IAS](#).
- Yao's lemma is all you need (for derandomization). [CMU Theory Lunch](#).
- Can we find an empty pigeonhole efficiently? [CMU Crypto Seminar](#).
- The Exact Complexity of Pseudorandom Functions and the Black-Box Natural Proof Barrier. STOC 2022 ([recorded full version talk on Youtube](#)).

Teaching Experiences

- Teaching Assistant: Spring 2023, *Fundamentals of Cryptography* at Tsinghua University.
- Teaching Assistant: Spring 2025, *6.1400 Automata, Computability, and Complexity Theory* at MIT.
- Teaching Assistant: Spring 2026, *18.405 Advanced Complexity Theory* at MIT.
- Organizer: Fall 2024, *Reading Group on Bounded Arithmetic* at MIT.